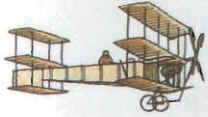


AIRCRAFT

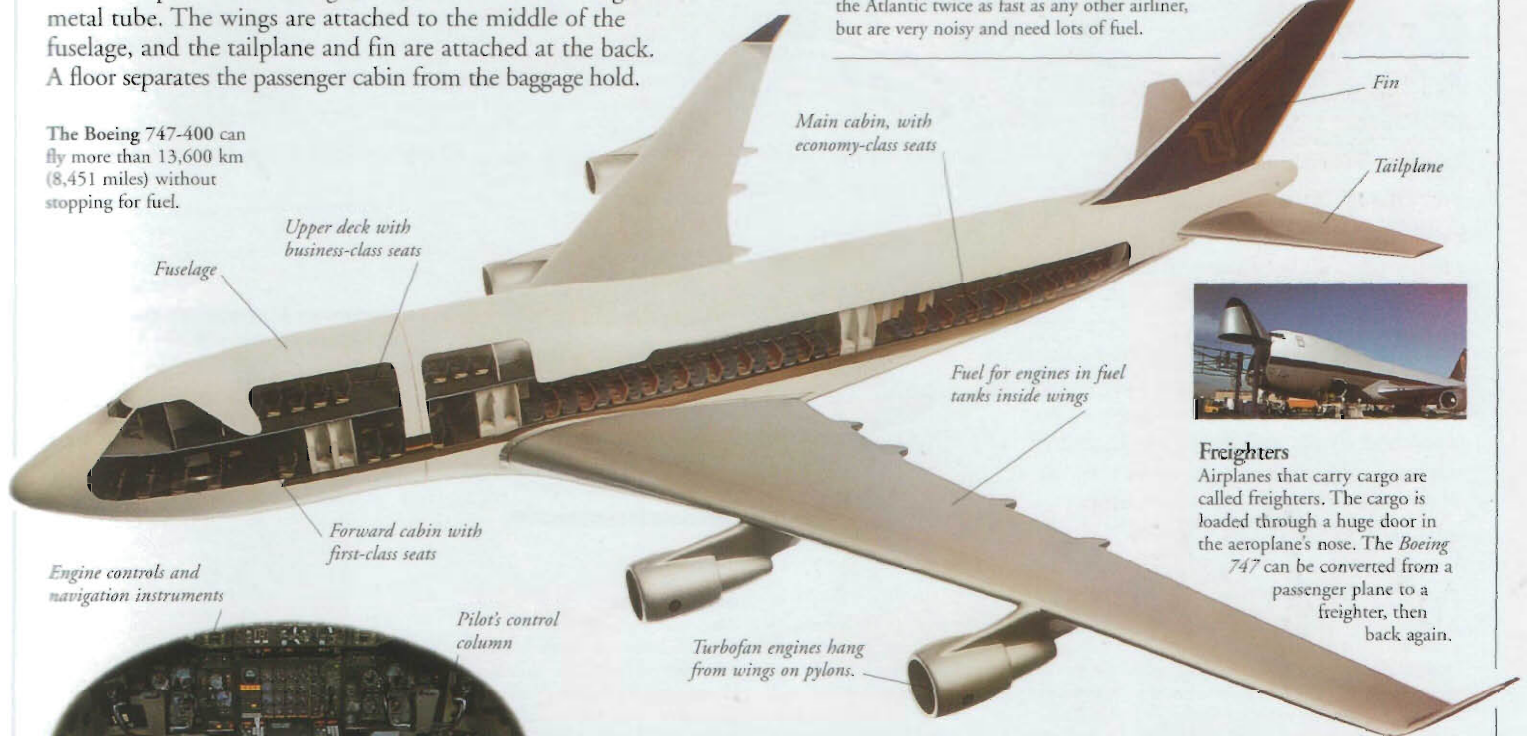


ANY VEHICLE THAT travels through the air is called an aircraft. The ability to soar over obstacles such as oceans and mountains makes aircraft the fastest form of travel. An airliner (a large passenger plane) can fly a passenger thousands of kilometres in hours. The same journey would take several days by boat or car. Airliners and military aircraft are complex machines. Their frames are built with lightweight metals, such as aluminium, and hi-tech materials, such as plastics. Inside, their sophisticated electronic controls help pilots fly efficiently and safely. Smaller aircraft, such as gliders and hot-air balloons, are often used for sport and leisure.

Anatomy of an airliner

Most airliners, such as this *Boeing 747-400*, have the same basic design. The main part is the fuselage, which is similar to a long, thin, metal tube. The wings are attached to the middle of the fuselage, and the tailplane and fin are attached at the back. A floor separates the passenger cabin from the baggage hold.

The *Boeing 747-400* can fly more than 13,600 km (8,451 miles) without stopping for fuel.



Cockpit

The aircraft is controlled from the cockpit. The pilot and co-pilot fly the plane using control columns, and instruments show the status of all the plane's equipment. The cockpit also contains radar and radio controls.

In-flight food

Pre-prepared meals are stored in trolleys, which lock into spaces in the aircraft's galleys until it is time for the cabin staff to serve them.



Entertainment

Some airliners feature video screens and headphones that can be tuned to music channels.

Howard Hughes

Hughes (1905–76) was an American industrialist, film-maker, and aviation enthusiast. He founded the airline TWA, and broke a number of aviation records in aircraft of his own design. Not all were successful; the *Spruce Goose* (1947) only flew once.



Types of aircraft

The word aircraft covers all flying machines – from balloons to helicopters. Most aircraft are aeroplanes, which have wings, and jet engines to give them speed. Other types of aircraft are gliders, which have no engines, helicopters, balloons, and airships. An aircraft's function determines its size and shape.

Biplanes

Many planes before World War II (1939–1945) had two pairs of wings, and were called biplanes.



Transport aircraft

Armies need aircraft to transport troops and equipment. Special aircraft are designed to carry very heavy objects, such as tanks.



Balloons

Lighter-than-air craft are known as balloons. A bag is filled with gas or hot air that is lighter than the atmosphere.



Glider

Currents of air move up and down. A glider has no engine, but flies by the effects of air currents on its wings.



Concorde

Supersonic airliners such as Concorde can travel faster than the speed of sound – about 1,240 kmh (770 mph). They can cross the Atlantic twice as fast as any other airliner, but are very noisy and need lots of fuel.



Freighters

Airplanes that carry cargo are called freighters. The cargo is loaded through a huge door in the aeroplane's nose. The *Boeing 747* can be converted from a passenger plane to a freighter, then back again.